

TOVA Eviction Decomposed — Four Independent Axes

v1 / EndurKV-Evict occupies the BUDGET axis only. Three other axes are open novelty space.

1. SIGNAL	2. SELECTION	3. BUDGET	4. INTER-HEAD
feature per position	top-K rule	K_h per head	coupling between heads
TOVA default - a[h, i] = current attn (plain TOVA scoring)	TOVA default - argmax-K of score (hard top-K)	TOVA default - K_h = K (fixed for all) (no per-head adaptation)	TOVA default - each head decides alone (no information exchange)
open ← we go next - Smoothed attention (heat-equation diffusion) - Spectral FFT energy (low-freq persistence) - K-graph centrality (PageRank on K-sim) - Cross-step variance⁻¹ - Cross-head consensus vote count	open ← we go next - Threshold-based (variable count / head) - Wave-propagated argmax (diffuse score → select) - Gumbel-soft sampling (stochastic top-K) - Graph-connected subgraph (k-core selection) - Compressed-sensing reconstruction	✓ explored (this paper) ✓ v1 (linear gate, α=1.3, β=0.6) final headline policy ✓ swept 69 gate shapes × 25 (α, β) × 7 thresholds ✓ signal/selection axes: negative result — TOVA's max_a + argmax is optimal - Result on Llama (720 cells): median −11% Δ KL vs TOVA 93% per-cell wins cachex = 1.02 (neutral)	open ← we go next - Cross-head consensus (k-core graph mechanism) - Mean attention across heads - Mean-field coupling - Eigenvector centrality across head-similarity - Shared budget pool within each layer

Composition strategy: best signal × best selection × best budget × best inter-head = full novelty stack

This composed policy is the base for Pivot A — thermal-coupled adaptive cache (α, β modulated by junction temp + battery SoC, on real phone silicon).